

Learning Objectives

- Apply differentiation to connected rates of change, small increments and approximations.

List of Terms

Constant Rate

恒定速率

A rate of change that remains the same over time. For example, traveling at a constant speed of 60 km/h.

Variable Rate of Change

变化率可变

A rate of change that varies over time. For instance, acceleration in a moving vehicle as it speeds up or slows down.

Related Rates of Change

相关变化率

Rates of change of two or more quantities that are connected by a relationship. For example, the relationship between the rate of change of the radius and the volume of an inflating balloon.

Small Changes

微小变化

Infinitesimally small variations in a quantity, often used in calculus to approximate changes using derivatives.

Important Notes

Rate of change	Related rate of change
The rate of change of a variable x with respect to time t is $\frac{dx}{dt}$.	In general, if two variables x and y are related by the equation $y = f(x)$, then the rates of change $\frac{dy}{dt}$ and $\frac{dx}{dt}$ are related by $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$
Small change If $y = f(x)$ and a small change δx in x causes a small change δy in y, then $\frac{\delta y}{\delta x} \approx \frac{dy}{dx} \text{ and so } \delta y \approx \frac{dy}{dx} \delta x$	

Warm Up

Follow along with the teacher to complete the following warm-up activities.

1. The radius r (cm) of a circle changes with time t (seconds) and they are related by the equation $r = 0.5t + 1.5$

a) Find the **initial value** of r and the **rate of change** of r .

2. The volume, V litres of water in a tank after t seconds is given by $V = 5 - \frac{2}{t+1}$

a) What is the **initial volume** of water?

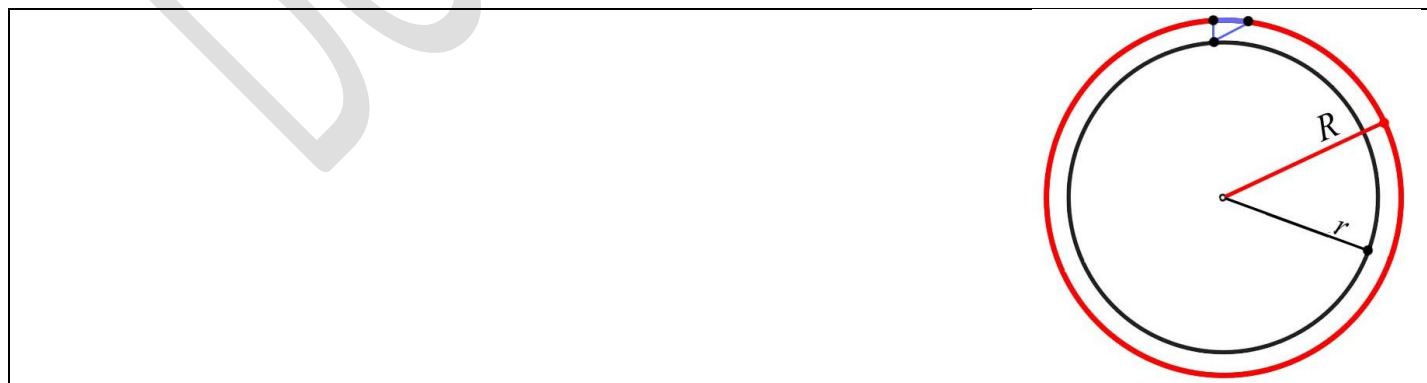
b) Find the **rate** at which the volume of water is increasing when $t = 3$



3. The radius of a circle **increases** at a rate of 3 cm s^{-1} . Find the **rate of increase of the area** when

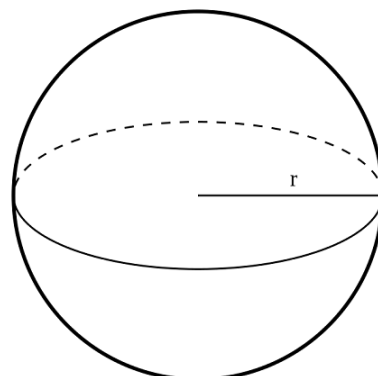
a) The radius is 5 cm

b) The area is $4\pi \text{ cm}^2$



4. Find, to 2 significant figures, the **rate of change** of the radius of a sphere if:

- the volume is **increasing** at a rate of $30 \text{ cm}^3/\text{s}$ when $r = 3$
- the volume is **decreasing** at a rate of $20 \text{ cm}^3/\text{s}$ when $r = 4$



5. Given $y = x^3$, find

- The approximate change in y if x increases from 2 to 2.01
- The approximate change in x if y decreases from 27 to 26.73

6. Given that $y = \frac{1}{2x+1}$, find the value of $\frac{dy}{dx}$ when $x = 2$. Hence find an expression in terms of p , for the approximate increase in y when x decreases from 2 to $2 - p$, where p is small.